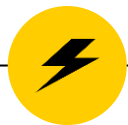


Computer Security Capstone

Supplementary: 5G Security



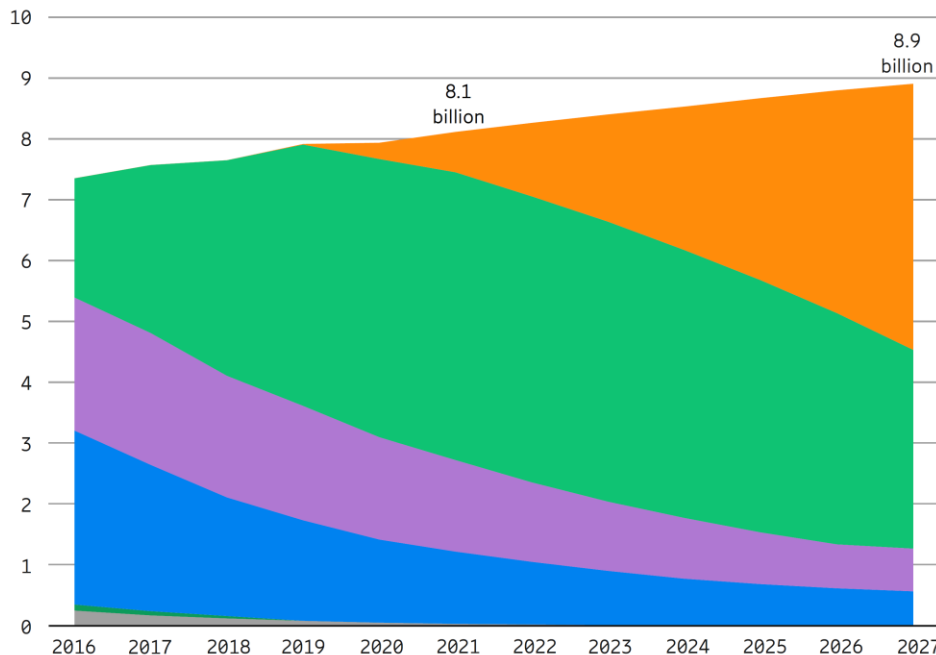
Computer Security Capstone

Chi-Yu Li

National Yang Ming Chiao Tung University

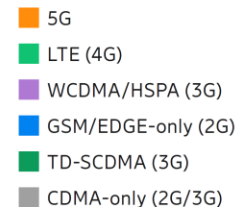


Mobile Subscriptions



5G subscriptions are forecast to reach 4.4 billion in 2027.

4.4bn



¹ GSA (October 2021).

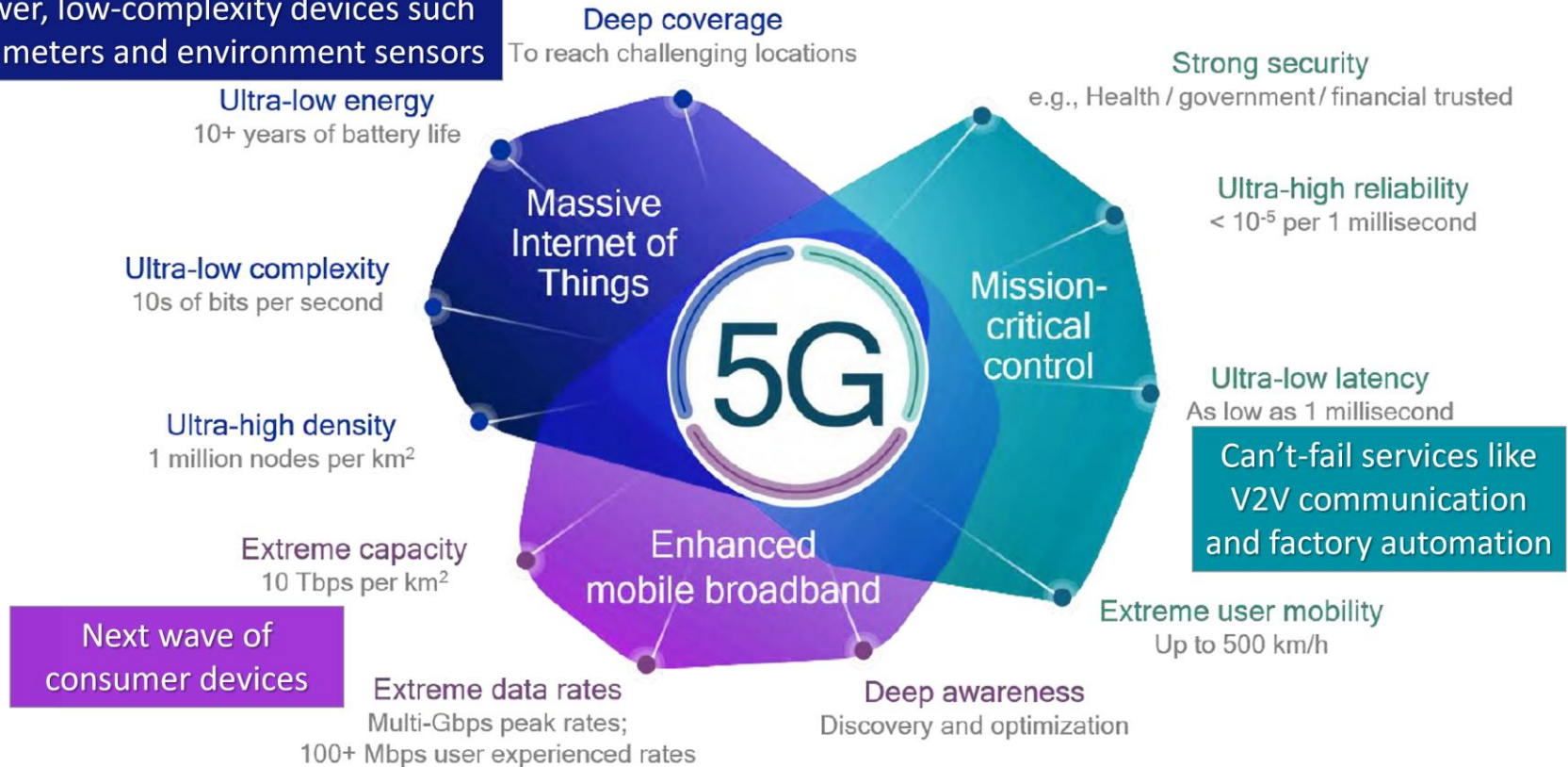
² A 5G subscription is counted as such when associated with a device that supports New Radio (NR), as specified in 3GPP Release 15, and is connected to a 5G-enabled network.

³ Mainly CDMA2000 EVDO, TD-SCDMA and Mobile WiMAX.



Various Requirements to 5G

Low-power, low-complexity devices such as smart meters and environment sensors





How to Deliver eMBB?

- Needs: higher throughput, lower latency, greater capacity, better uniformity and complete mobility
- Communication technologies
 - Massive MIMO
 - More spectrum sharing
 - mmWave
 - Gigabit LTE
 - Device-centric mobility



How to Deliver MCC?

- Needs: faster than humans can think; failure is not an option
- Enhanced ultra-reliable, low-latency communication (eURLLC)
 - Scalable slot duration down to 125 μ s
 - Efficient multiplexing with scheduled traffic
 - Spectrum sharing allows for more-predictable QoS
 - Redundant links to mission-critical devices with multi-connectivity



How to Deliver Massive IoT?

- ◎ Needs: more efficiently connect the wide variety of IoT devices and services

- ◎ Communication technologies
 - Upon the foundation of NB-IoT
 - More efficient uplink transmission scheme for IoT with RSMA
 - WAN-managed multi-hop mesh to extend network coverage



5G Revolution

- ◎ Service-based Core: delivering multi-network slicing, multi-level of services and multi-connectivity network capabilities
 - Based on virtualization and SDN
 - flexibility, agility and economics of scale
- ◎ Addressing many threats faced in today's 4G/3G/2G
 - E.g., new mutual authentication capabilities
- ◎ However, adopting new network technologies introduces new potential threats
 - Increasing attack surface



Security in 5G: Threats Everywhere

5G Devices

RAN

Edge Network

5G Core Network

Internet

Internet Threats

Device Threats

- Bots
- DDoS
- MitM Attacks
- Firmware Hacks
- Device Tempering
- Malware
- Sensor Susceptibility

Air Interface Threats

- Jamming
- MitM Attacks
- Eavesdropping

Edge Network Threats

- MEC server vulnerabilities
- Rogue Nodes
- Authentication Issues
- Side Channel Attacks
- Improper Access Control

Backhaul Threats

- DDoS Attacks
- Control and user plane sniffing
- MEC Backhaul Sniffing
- Flow modification attacks

5G Core Network Threats

- Software issues
- API vulnerabilities
- Networking Slicing issues
- DoS and DDoS attacks
- Improper Access Control
- Virtualization issues



5G Security Highlights

- New attack surface: changes from legacy cellular
 - Inter-working between multiple technologies and multiple generations, even non-trusted environments
 - User device proliferation
 - Moving intelligence from the core to the edge of networks
 - Network virtualization
 - Shift from telecom network protocols to IP-based protocols
 - Convergence of multiple technologies
 - More open platform/technology stacks with software from more vendors



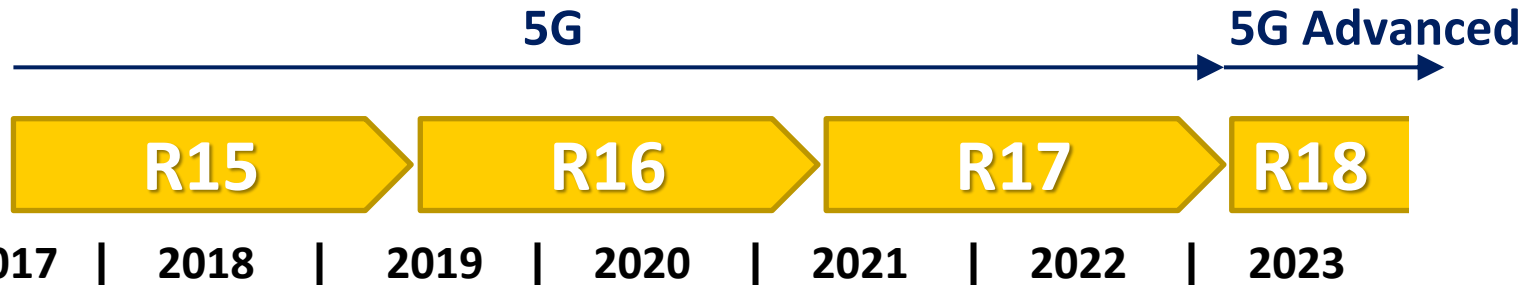
5G Security Highlights (cont.)

- ◎ New attack surface: new innovations and verticals
 - Cloud RAN, vRAN, O-RAN
 - Service based core network architecture
 - SDN, NFV, network slicing, etc.
 - Cloud and edge computing
 - Limits of standards-specified protection



5G Security Evolution

- Main document: 3GPP TS 33.501
 - Security architecture and procedures for 5G system



Security for **vulnerability resolution** and **new architecture/functions**

Security for **new essential functions/services**

Security for **edge and management functions**

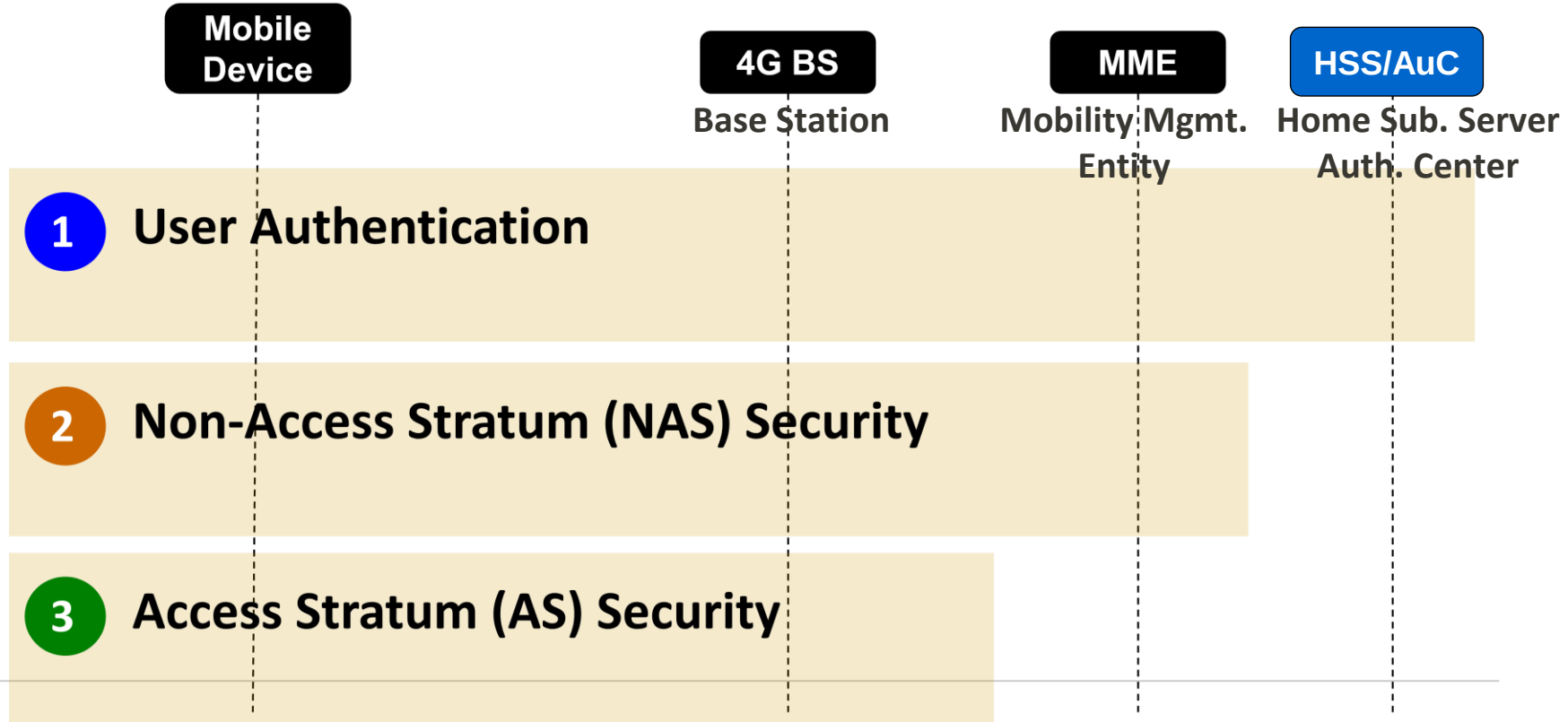


5G Security in 3GPP R15

- ◎ Security for vulnerability resolution from legacy security
 - **Subscriber ID privacy:** ID is never disclosed over the air
 - **Increased home control:** Home network makes final auth. decision
 - **Security edge protection proxy (SEPP):** security between two networks

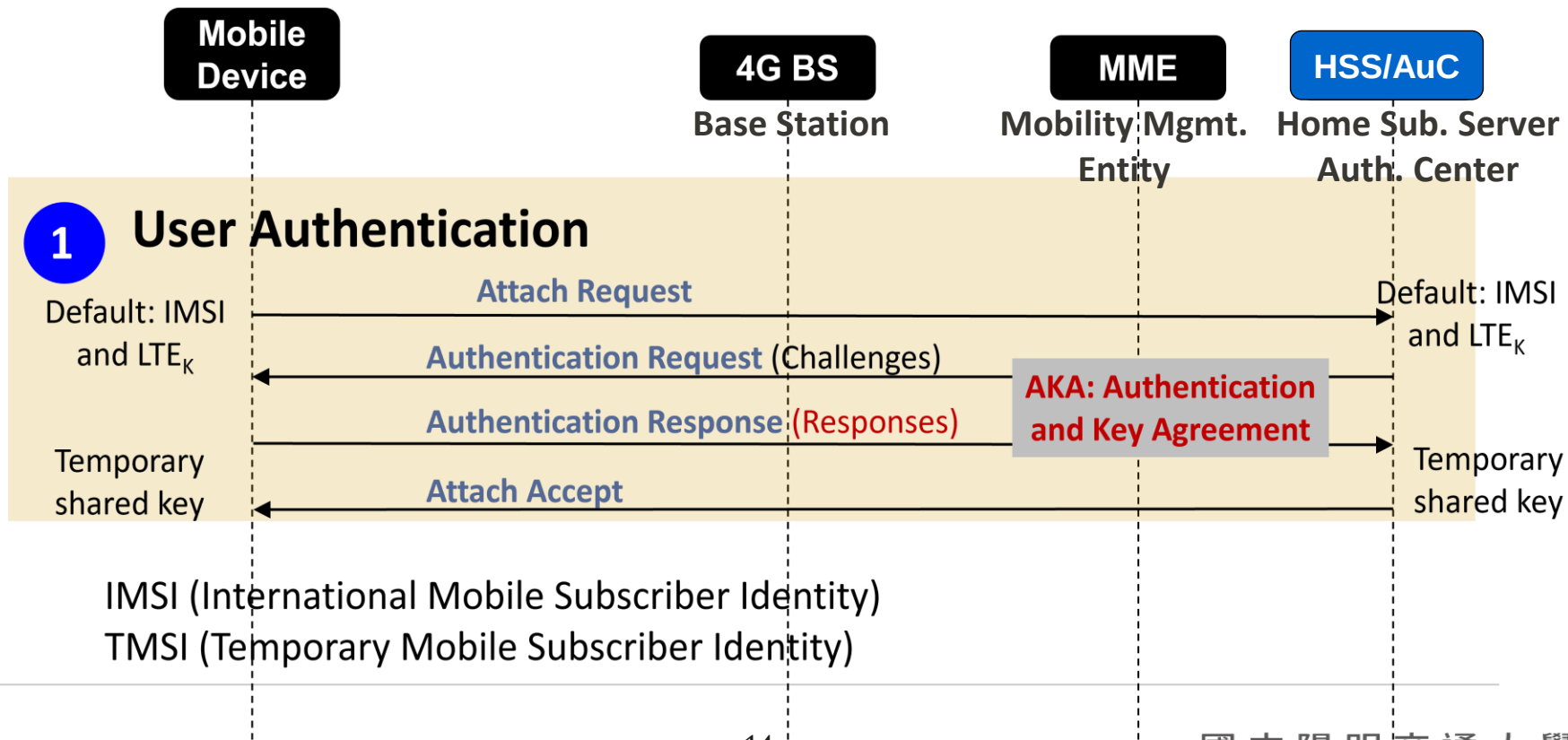


Legacy 4G Security



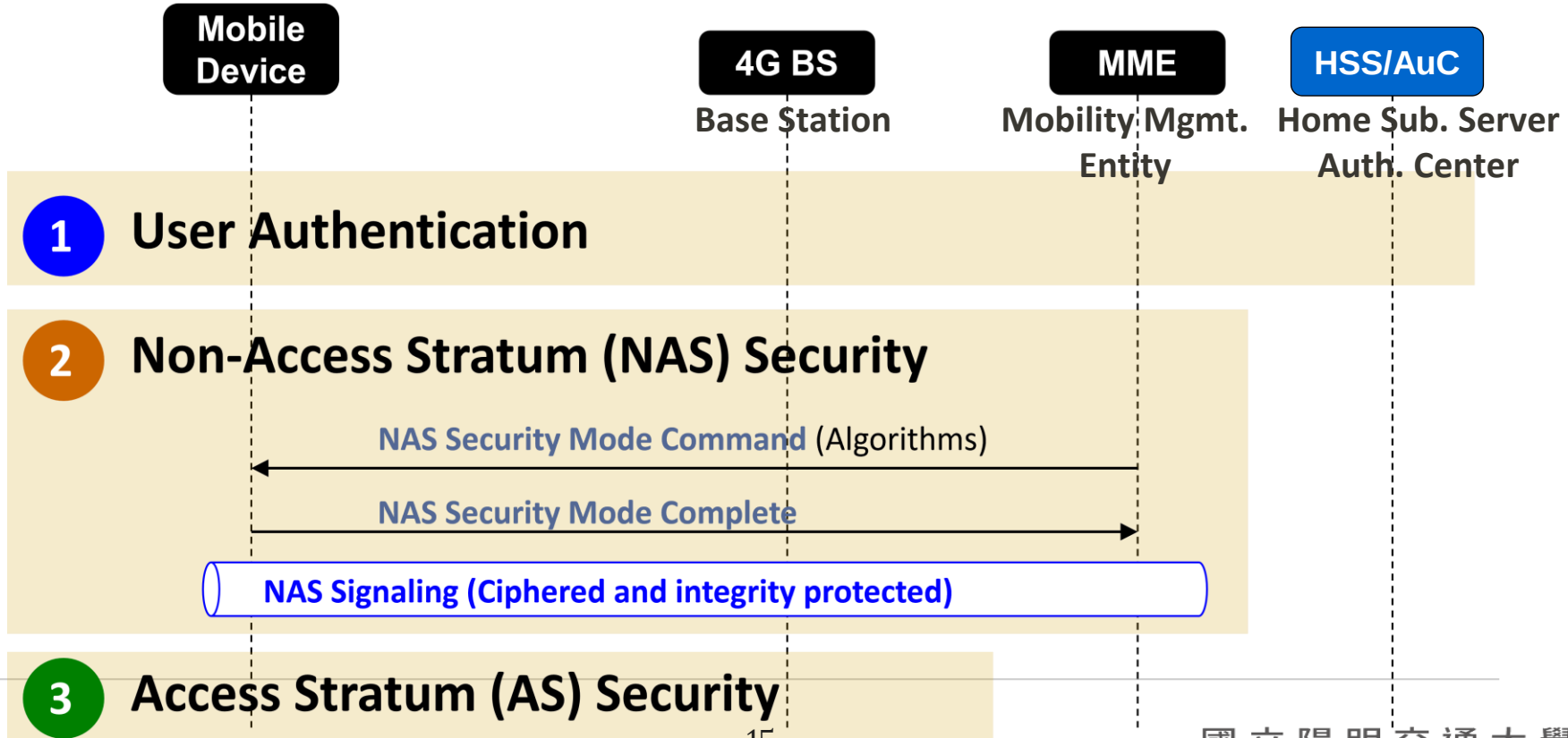


Legacy 4G Security (Cont.)



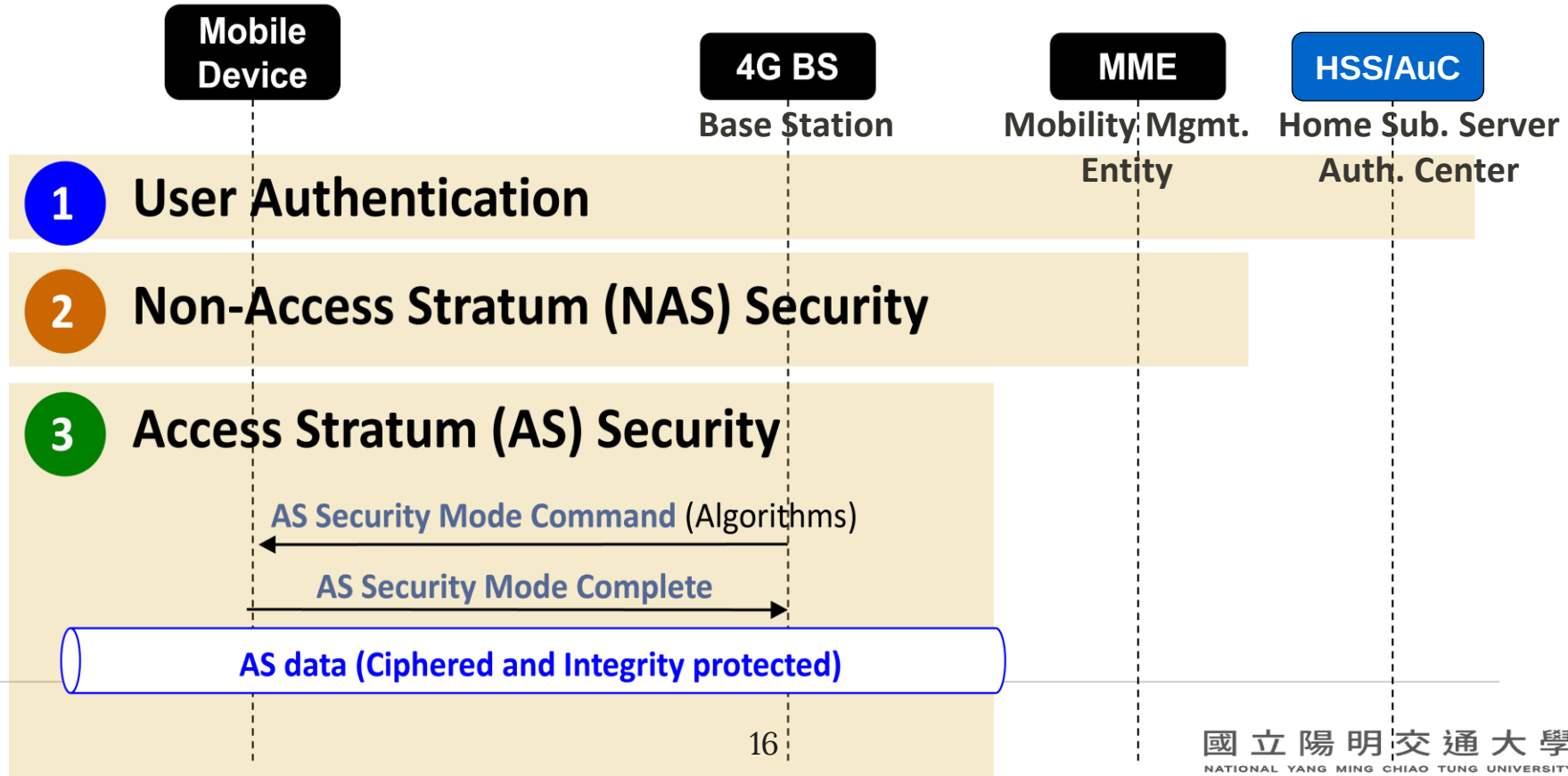


Legacy 4G Security (Cont.)





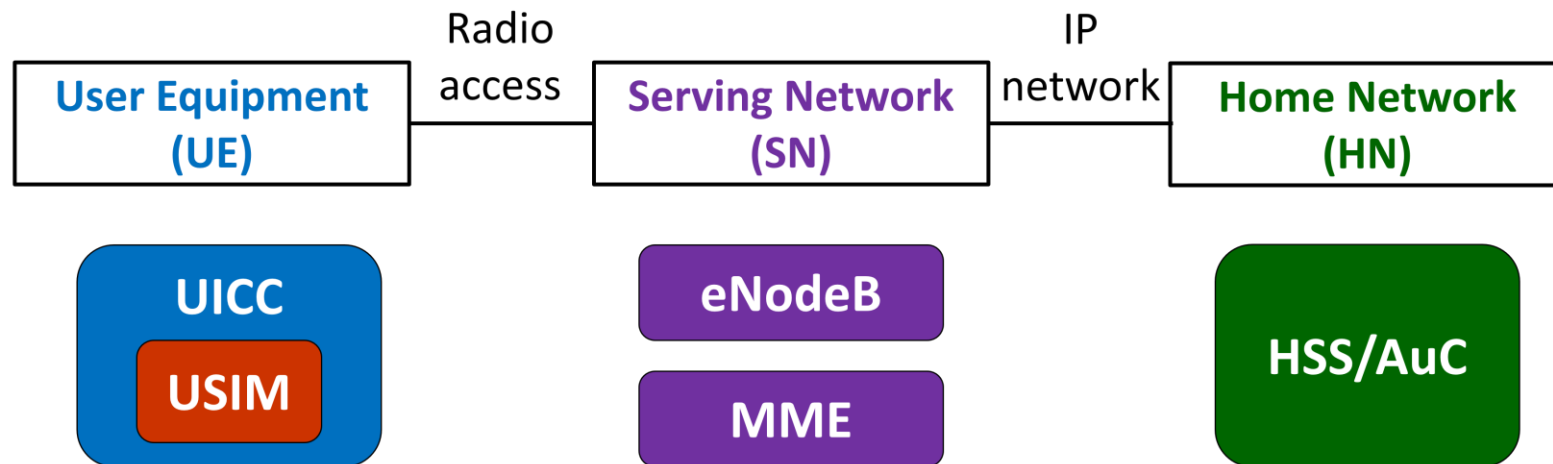
Legacy 4G Security (Cont.)





4G Authentication Framework

● EPS-AKA: EPS-Authentication and Key Agreement



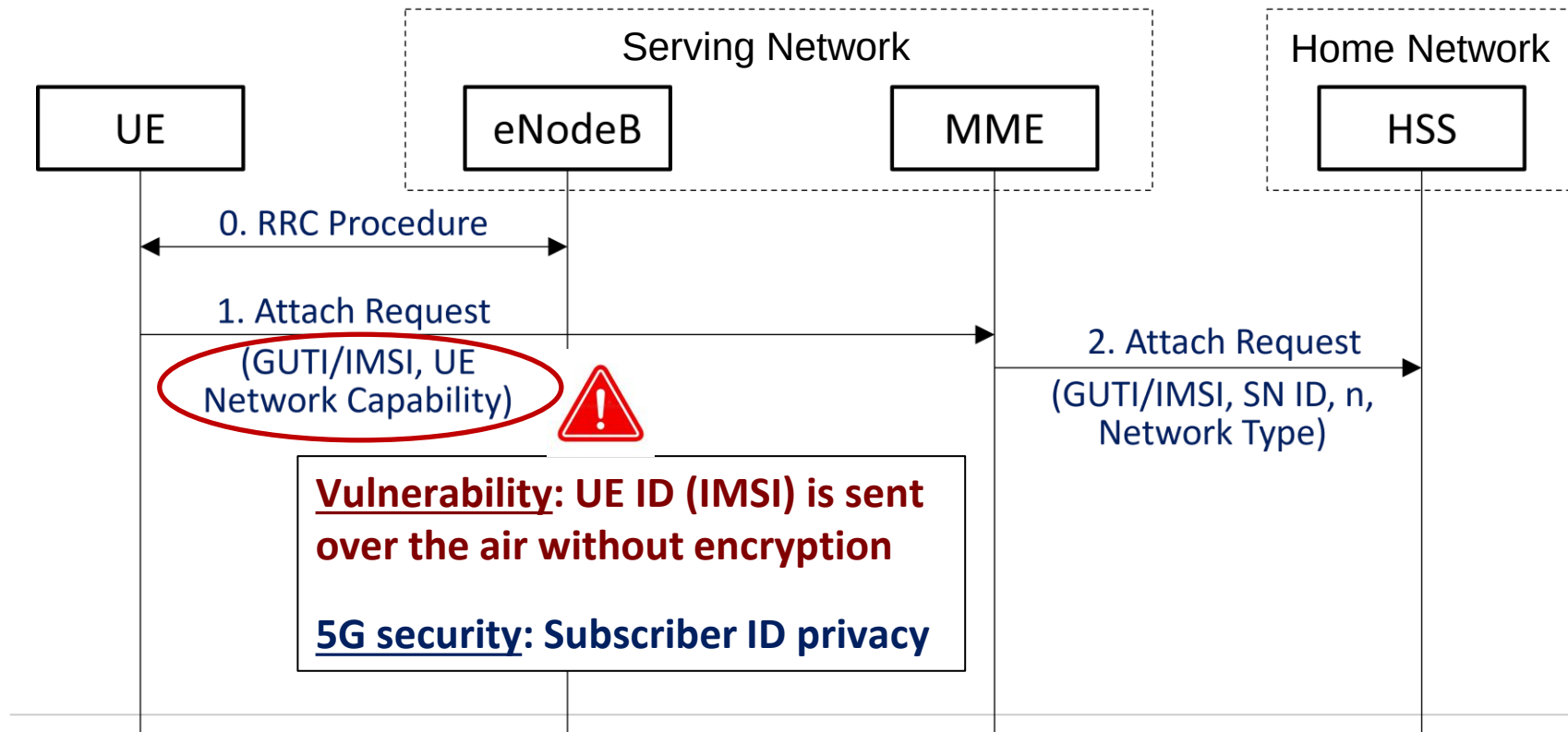
UICC (Universal Integrated Circuit Card)
USIM (Universal Subscriber Identity Module)
- storing a cryptographic key shared with the HN

eNodeB (Evolved NodeB)
MME (Mobility Management Entity)

HSS (Home Subscriber Server)
- Storing user credentials
- Authenticating UE

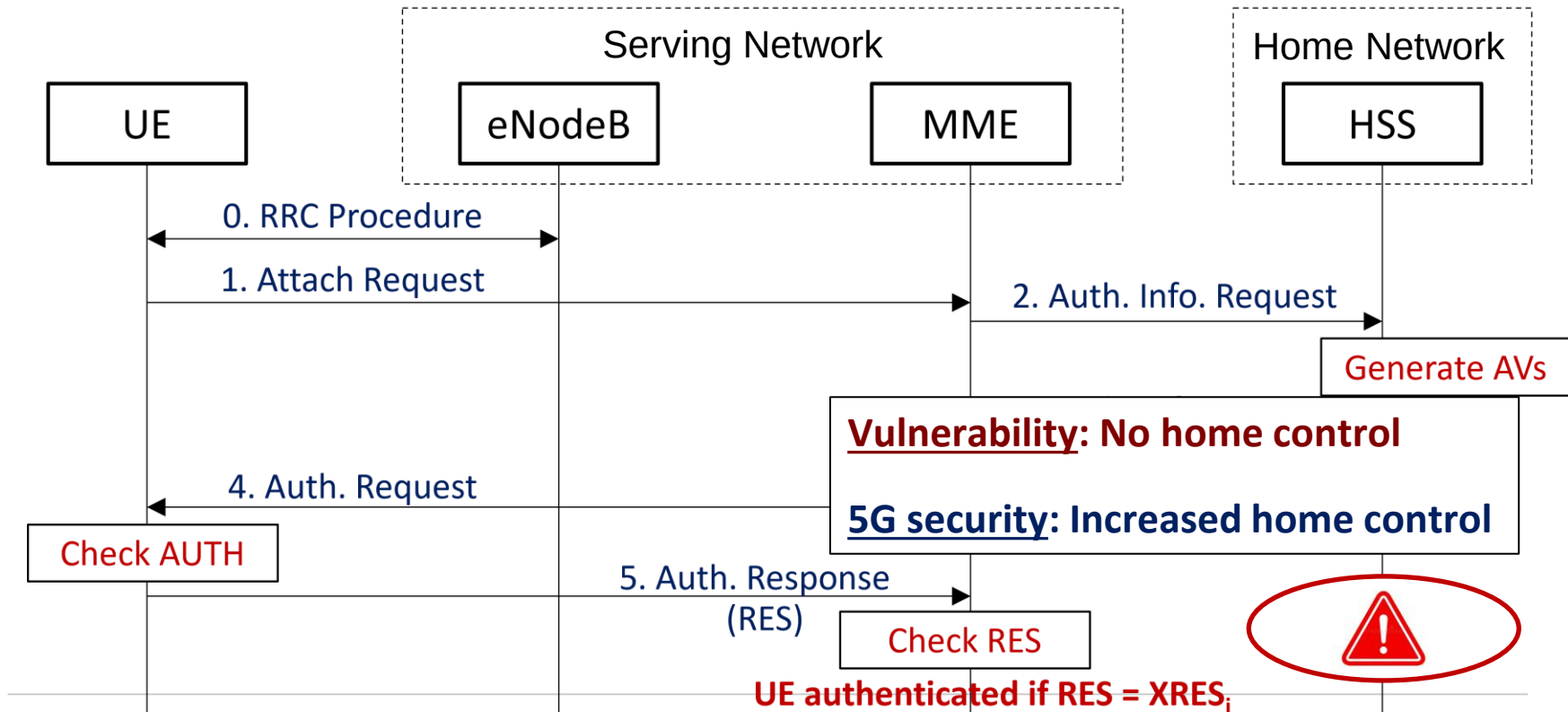


4G EPS-AKA Procedure



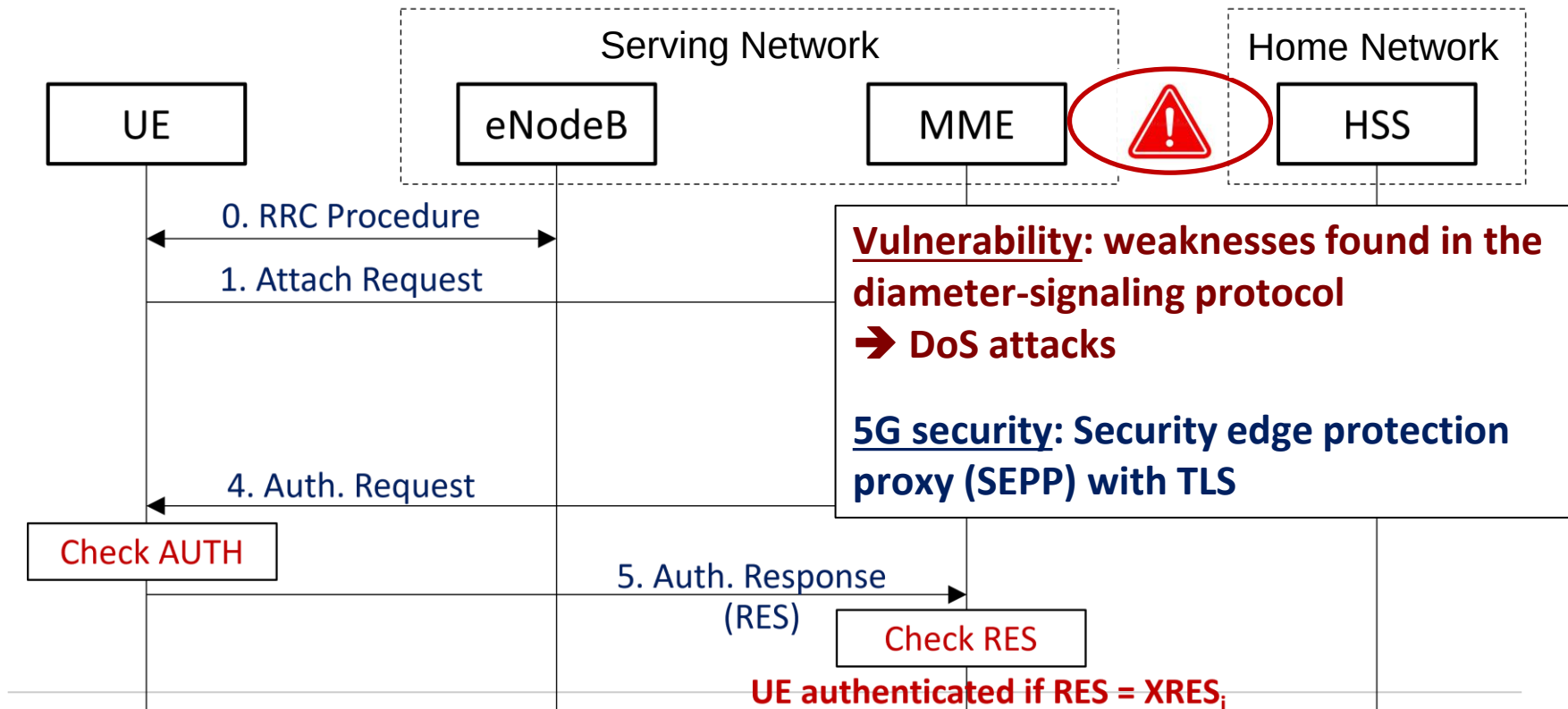


4G EPS-AKA Procedure (cont.)





4G EPS-AKA Procedure (cont.)





5G Security in 3GPP R15

- ◎ Security for vulnerability resolution from legacy security
 - **Subscriber ID privacy:** ID is never disclosed over the air
 - **Increased home control:** Home network makes final auth. decision
 - **Security edge protection proxy (SEPP):** security between two networks

- ◎ Security for new architecture and functions
 - **Unified authentication framework:** 3GPP and non-3GPP access networks
 - **Service based architecture (SBA) security**

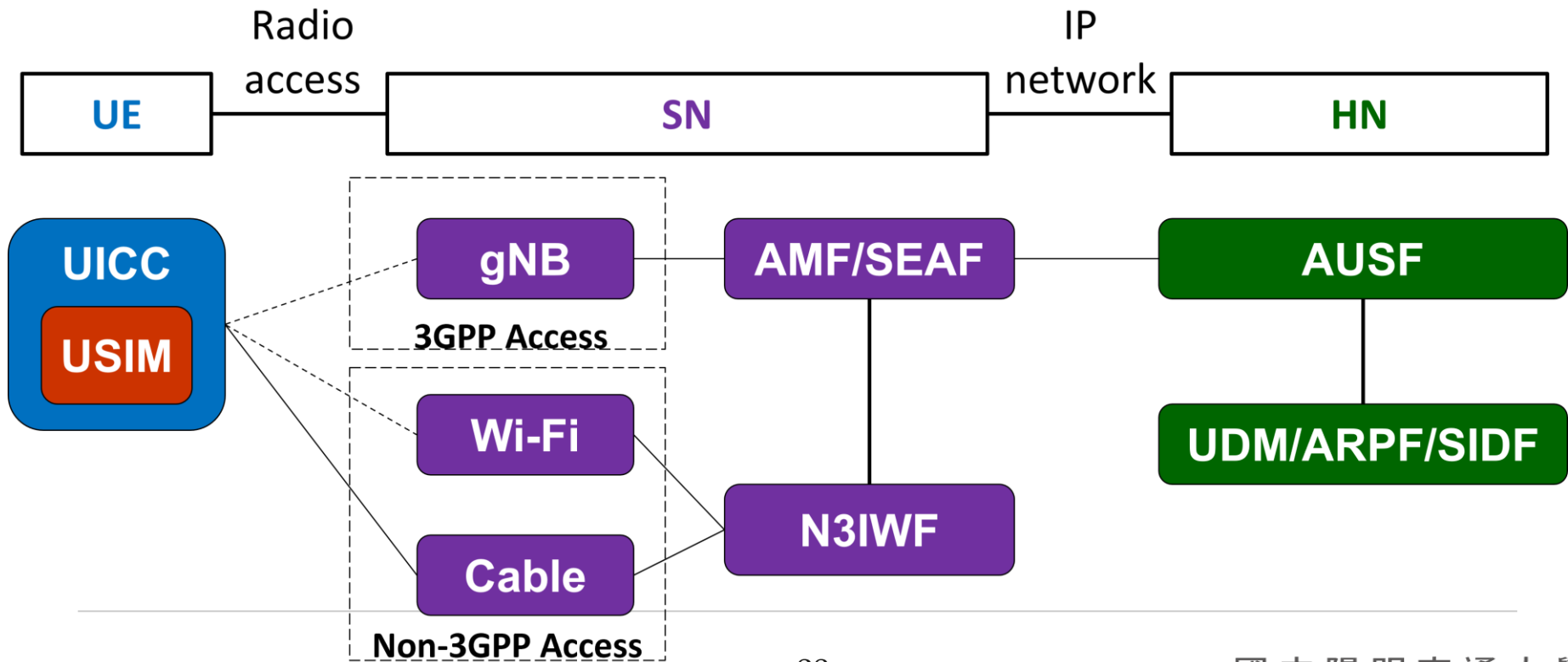


5G Unified Auth. Framework

- ◎ Three authentication methods
 - **5G-AKA & EAP-AKA'**
 - Trust model: shared symmetric key
 - **EAP-TLS**
 - Limited use cases: private networks and IoT environments
 - Trust model: public key certificate
- ◎ Why EAP (Extensible Authentication Protocol)?
 - Allowing the use of different types of credentials besides the ones commonly used in mobile networks



5G Unified Auth. Framework



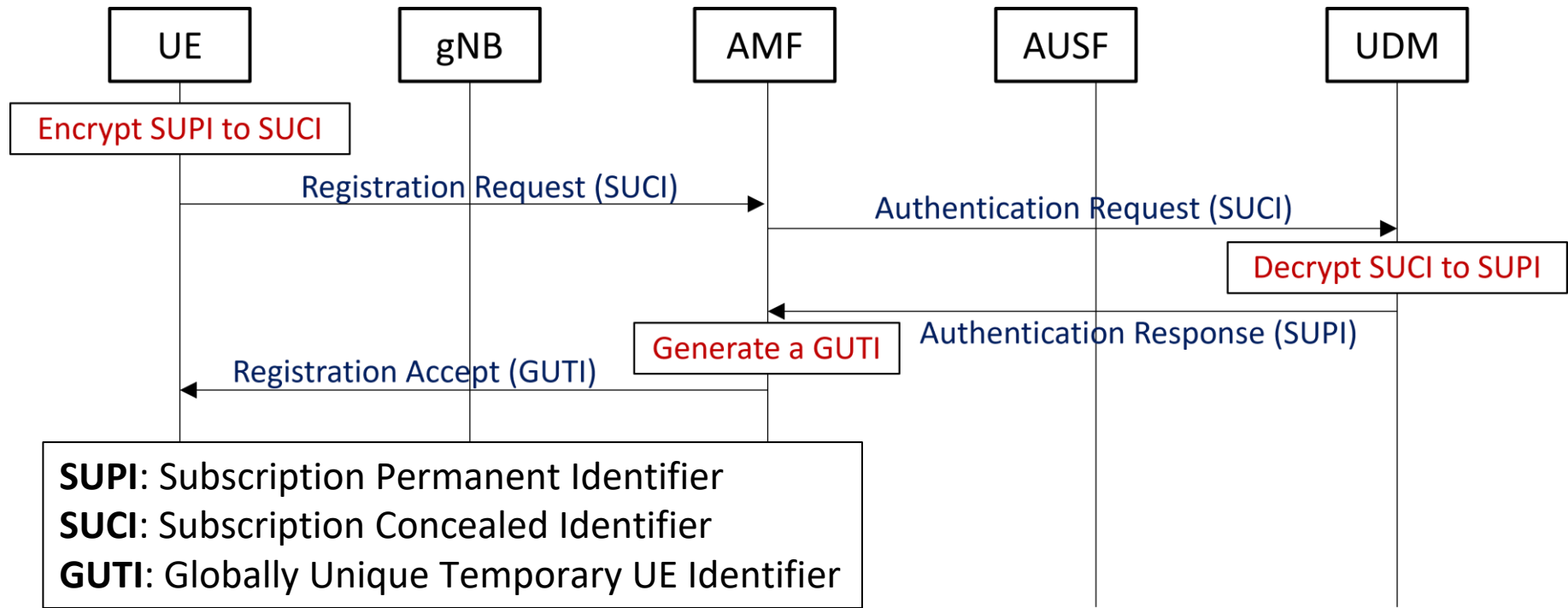


5G Subscriber ID Privacy

- ◎ **SUPI**: Subscription Permanent Identifier
- ◎ **SUCI**: Subscription Concealed Identifier
- ◎ **GUTI**: Globally Unique Temporary UE Identifier

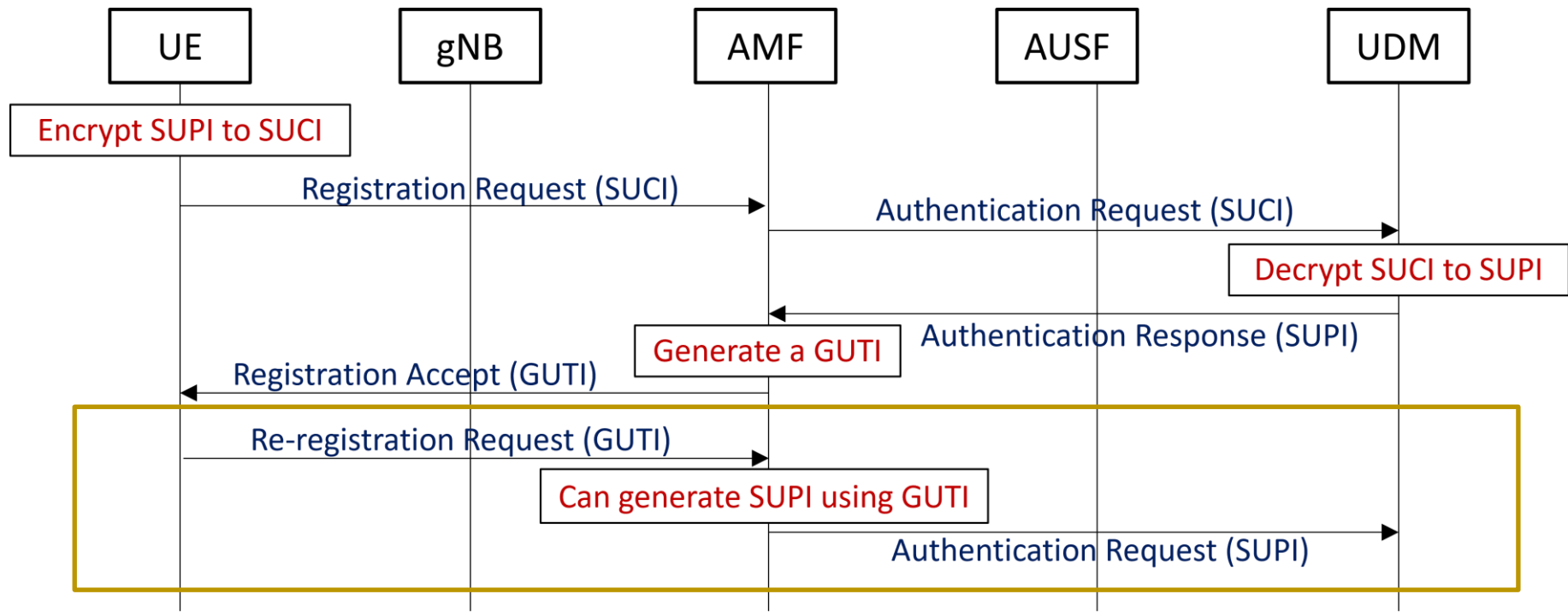


5G Subscriber ID Privacy



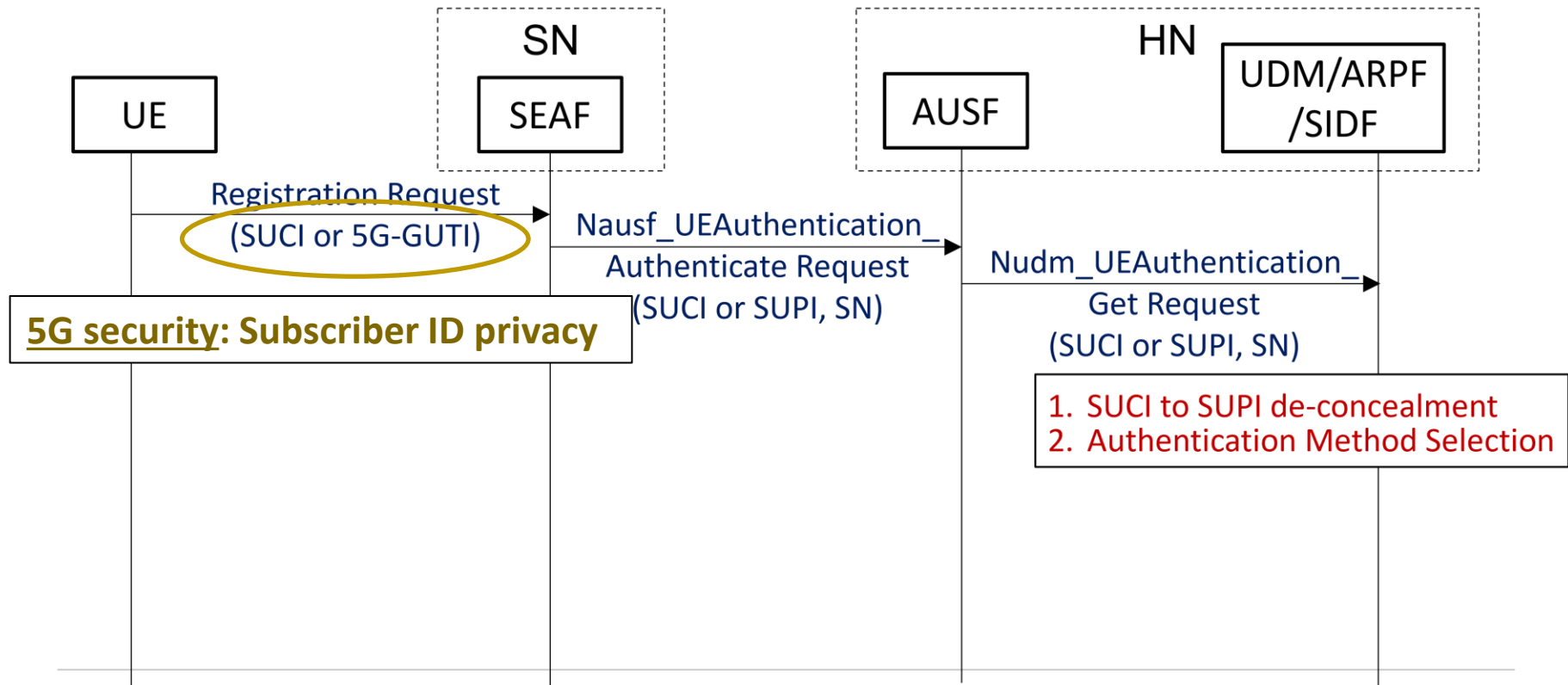


5G Subscriber ID Privacy



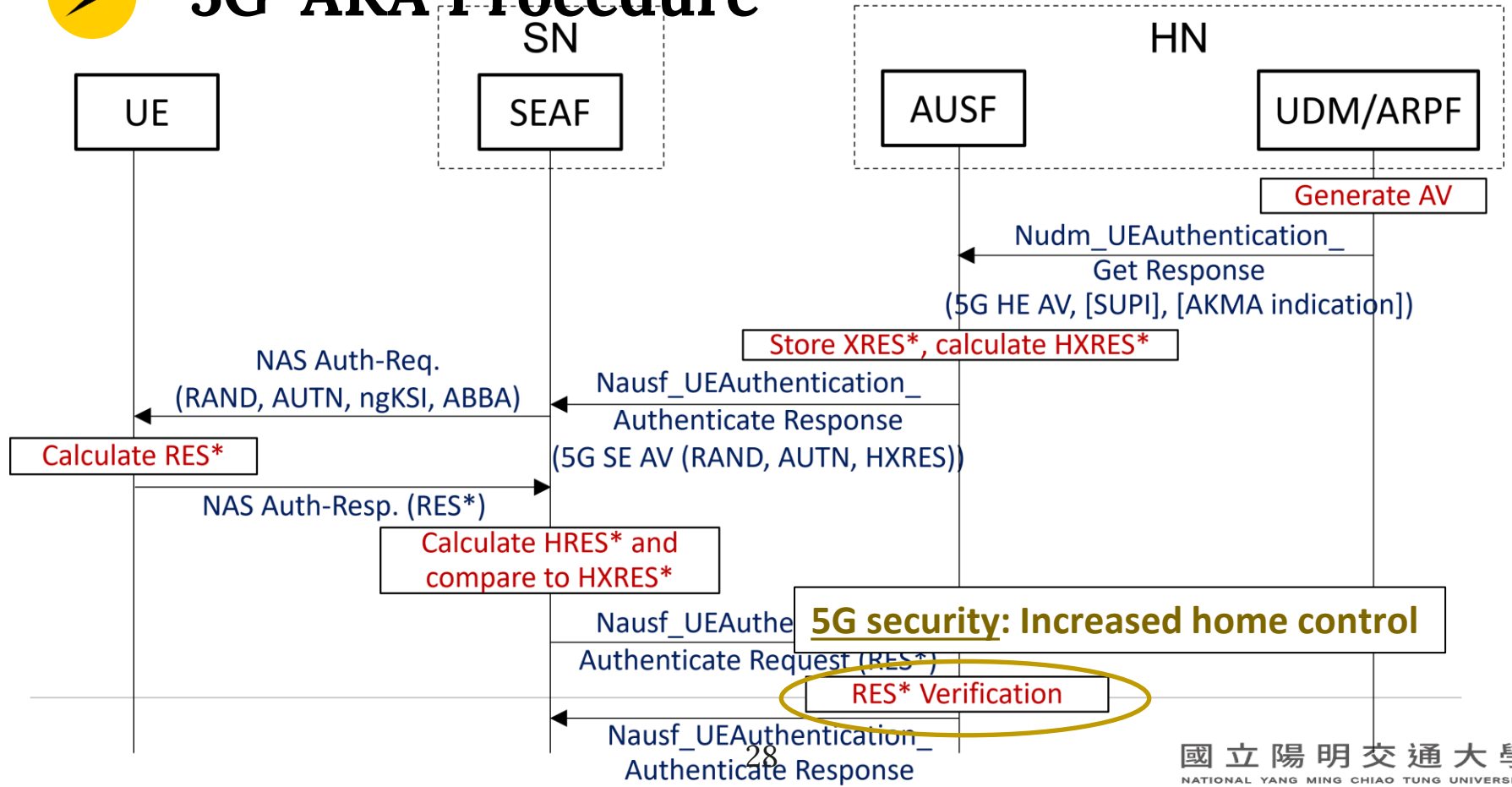


5G-AKA Procedure





5G-AKA Procedure





5G Security in 3GPP R15

- ◎ Security for vulnerability resolution from legacy security
 - **Subscriber ID privacy:** ID is never disclosed over the air
 - **Increased home control:** Home network makes final auth. decision
 - **Security edge protection proxy (SEPP):** security between two networks

- ◎ Security for new architecture and functions
 - **Unified authentication framework:** 3GPP and non-3GPP access networks
 - **Service based architecture (SBA) security**



5G Security in 3GPP R15

- Security for vulnerability resolution from legacy security
 - Subscriber ID privacy: ID is never disclosed over the air
 - Increased home control: Home network makes final auth. decision
 - Security edge protection proxy (SEPP): security between two networks
- Security for new architecture and functions
 - Unified authentication framework: 3GPP and non-3GPP access networks
 - Service based architecture (SBA) security





Why 5G SBA?

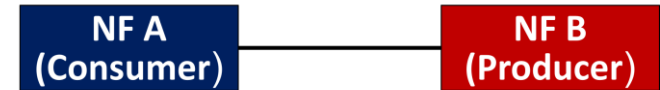
- **Updating production network:** loosely-coupled services
- **Extensibility:** light-weighted service-based interface
- **Modularity and Reusability:** easily invoked by other services
- **Openness:** easily exposed to external users



5G SBA: NFs Security

- ◎ Direct communication
 - Transport-layer protection (e.g., TLS)
 - Token-based authorization (OAuth 2.0)

- ◎ Indirection communication via SCP
 - Implicit authentication
 - Token-based authorization (OAuth 2.0)



Discovery of NF B by local configuration or via NRF



Direct discovery or delegation of discovery to SCP



5G Security in 3GPP R16

- ◎ Security for new essential functions and services
 - Network slices
 - Non-3GPP access
 - Non-public network
 - Time Sensitive Communications (TSC) service
 - Integrated Access and Backhaul (IAB)
 - Ultra-Reliable and Low Latency Communications (URLLC) service



5G Security in 3GPP R17

- Security for edge and management functions
 - Edge computing
 - Multicast/broadcast service
 - Message service for Massive IoT (MIoT)
 - Network Data Analytics Function (NWDAF)



Certifications/Audit Enhancement

- ◎ NESAS (Network Equipment Security Assurance Scheme) jointly defined by GSMA and 3GPP
 - Security evaluation of mobile network equipment
- ◎ Benefits for vendors
 - Accreditation from the world's leading mobile industry representative body
 - Offers a uniform approach to security audits
 - Avoiding fragmentation of requirements in different markets
- ◎ Benefits for operators
 - Rigorous security standard requiring a high level of vendor commitment
 - Peace of mind for appropriate security measures and practices
 - No need to spend money and time conducting individual vendor audits



Conclusion

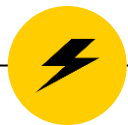
- ◎ 5G security designs from 3GPP standard
 - (R15) Security for **vulnerability resolution** and **new architecture/functions**
 - (R16) Security for **new essential functions** and **services**
 - (R17) Security for **edge** and **management functions**
- ◎ NESAS: Security for mobile network equipment
- ◎ **However, any of design flaws, implementation bugs, and operation slips may cause security issues to a system**
 - Especially for new architecture/functions/services



Conclusion (cont.)

- So, more security efforts are required beyond the designs
 - Threat prevention
 - Anomaly detection
 - Attack response
 - Loss recovery

Thanks for Your Attention!



Computer Security Capstone



Reference

- 3GPP TS 23.501, “System architecture for the 5G System (5GS).”
- 3GPP TS 33.501, “Security architecture and procedures for 5G system.”
- 3GPP TS 33.535, “Authentication and Key Management for Applications (AKMA) based on 3GPP credentials in the 5G System (5GS).”
- 3GPP TS 33.813, “Study on security aspects of network slicing enhancement.”
- “Ericsson Mobility Report,” Ericsson, Nov. 2021
- Ana Schafer, “Enhanced Mobile Broadband – 5G Innovation for consumers?” Qualcomm developer network, 2019
- “A Comparative Introduction to 4G and 5G Authentication,” CableLabs, 2019.
- “5G Security when Roaming - Part 2,” Mpirical, 2022. [Online]. Available: <https://mpirical.com/blog/5g-security-when-roaming-part-2>
- IETF RFC 6749, “The OAuth 2.0 Authorization Framework.”